Netlet — Requirements Document (Developer Version)

(نسخة المطورين) وثيقة المتطلبات — نت ليت) — Version date: August 11, 2026

1. Overview

Netlet is a multi-vendor marketplace similar to Amazon and Noon, being rebuilt completely from scratch after the current version faced major technical problems (frequent bugs, an outdated architecture that is hard to develop further, and weak backend/frontend design). The goal is a powerful, modern platform built by a local development team, capable of competing with the major players in the Gulf market.

2. General Architectural Principles

- Fully microservices-based architecture: any new feature or service added in the future is built as an independent microservice, without affecting the rest of the system.
- Easy-to-use experience for all parties: the vendor, the buyer, the admin/management team, the delivery company, and the employee processing orders.
- Full flexibility: every advanced feature is optional (can be enabled, or left on its simple default behavior) — the system is never rigid or locked into one single mode.
- AI is a core part of the architecture, not an afterthought — especially in customer service and product management.

3. Phase 1

3.1 Customer Experience

- Request a notification when an out-of-stock product becomes available again — this request also reaches the vendor, so they know the level of demand for their product.
- Optional installation fees and extended warranty on eligible products.
- Pre-order products that ship once available.
- Public and private Wishlists — friends and family can enter the customer's phone number to see the public list and buy an item from it as a gift.
- AI-personalized recommendations based on each customer's interests.
- AI visual search: upload a photo of a product or describe it to get similar available products, with the same feature available directly from any product page.
- Smart, professional display of product variants (color, size) — changing the color swaps the displayed product photo, with zoom capability and attributes customized per product category.
- NEW: AI Virtual Try-On — for clothing, accessories, and jewelry: the customer can photograph themselves (or their hand) and see how the product would look on them before buying, reducing returns and increasing purchase confidence.
- Add videos to product pages alongside photos.
- AI automatically pulls the manufacturer's official product images (not AI-generated new-angle photos).
- Quick View on category/listing browsing pages: a tap or long-press on a product opens a popup with its details and the option to add to cart or buy directly by choosing a quantity, without opening the full product page.
- Icons alongside category names for easy visual recognition.
- A "request special discount" feature on a specific product, pending vendor approval (opt-in per vendor).
- Referral system with two tracks: (a) a specific product link — the commission applies only to that product; (b) a general invite — any first purchase by the invitee grants the referrer an ongoing commission. Referral commissions are only paid out after the legal return period has passed, to prevent abuse.
- A set percentage (admin-configurable) for the product-link referral commission, credited as points to the person who shared the link with their circle. The points holder can either spend them in full on purchases, or request a cash payout with a 10% cash-collection fee deducted from the value.
- Reviews: allowed only for a Verified Purchase, and only after delivery is actually confirmed — not just after purchase.

- Cart Recovery — if a customer adds items to the cart and doesn't complete the purchase, they receive a smart reminder after a set period (possibly with a small incentive like a discount code), to raise checkout completion rates.
- Price-drop alerts — a customer can "follow" a specific product, and if its price drops or a discount is applied, they get an instant notification.
- Subscribe & Save for recurring products — for consumable products the customer buys regularly, later linked to the Grocery section.
- NEW: Smart AI Size Guide — AI suggests the right size for the customer based on their past purchases or measurements, reducing returns caused by wrong-size selection.
- NEW: Frequently Bought Together — when a customer buys a specific product (e.g. a phone), complementary products appear (such as a case or screen protector), whether from the same vendor or from other vendors.

3.2 Checkout & Payments

- The platform's own payment service supporting: KNET, credit cards, Deema (installment/BNPL), and a direct payment link.
- Cash on Delivery (COD) available as a default option, with the ability to disable it administratively when needed.
- Additional payment options at checkout: points, gift card, discount codes.
- A shopping cart that supports products from multiple vendors in a single order, with a separate delivery cost per vendor.
- Choice of delivery speed (fast/regular) per individual product within the same order.
- Choice of a different delivery address per individual product within the same order.
- Address selection via a smart interactive map (pin a location on the map or add a new address through it).
- Ability to split a single order into independent sub-operations per product (purchase, return, delivery, re-pickup) — e.g. returning one product from a multi-item order without affecting the rest.
- Top priority given to checkout speed (Fast Checkout).

3.3 Vendor Experience

- Self-registration flow: (1) the user creates a basic account (username/password), (2) chooses to register as a customer or a vendor, (3) if "vendor" is chosen, they complete the company registration steps (required info and documents) within the same account and submit for review — they don't get access to the vendor dashboard at this stage, only filling in information, uploading documents, and waiting, (4) we respond either by requesting additional documents or by sending the final contract (currently as an attached file, with e-signature planned for the future) for them to sign and upload, (5) once approved, the vendor dashboard opens and they can start adding products. The same account is used as both customer and vendor, with the ability to switch between the two modes.
- The vendor's default commission rate is shown during onboarding, with the option to propose a custom rate or fixed amount for specific products, subject to later admin approval (included in the contract if approved).
- Adding products individually or in bulk via an Excel file or other means, with AI assistance for entering product data.
- AI automatically adjusts product images after upload: standardizing dimensions, improving quality, and detecting missing photos or required details — the same applies to bulk uploads.
- AI automatically categorizes products into the correct category during bulk upload.
- AI reviews and approves products after they're added, alerting the admin if any issue is detected before the product goes live.
- Some large/trusted companies are granted permission to add products directly, without going through the usual approval cycle.
- Any change a vendor makes to a product's price, or adding a new product, or deleting/removing a product, is immediately visible to the admin (notification + log) to track all changes to vendor products.

- Commission rates/values are entirely outside the vendor's control — they can only be edited by the admin, or by an employee specifically authorized to do so. The system logs who made the change (employee name) and which admin granted that employee this specific permission.
- Vendor Trust Badges displayed next to their name on the product page, such as: Verified Seller, Fast Shipping, High Rating — giving the buyer instant credibility.
- NEW: A vendor can build complementary product Bundles for their product — either from their own products, or in coordination with other vendors to feature their products as an add-on (e.g. a phone vendor linking a case or screen protector from another vendor).
- Fully flexible commission setup: percentage or fixed amount, per vendor company or per individual product.
- Per-vendor choice of delivery provider: their own delivery company or the platform's own delivery service, configurable from the system.
- When a product's size/type is entered, the delivery company automatically sees the required and available vehicle type, making it easier to accept the order and calculate the cost.
- Self-service booking for a featured product placement (e.g. within the top 5 results) or an ad banner, via a calendar showing available dates, price, and display method — vendors who don't pay appear in the normal listing.
- A dedicated path for individuals without a registered company: they sell under Netlet's own license, with sale proceeds going to Netlet first, then a weekly report in bank-transfer format to pay out the seller's dues after deducting the commission.
- Ability to switch the account between vendor mode and regular customer mode from the settings page.
- Automatic suspension of a vendor's account after 3 violation warnings.
- Requiring discount-license information to be attached whenever a vendor creates a discount offer (a legal requirement in Kuwait).
- Allowing vendors to participate in adding discounts to their products, so vendors who participate in discounts get priority or extra privileges (such as better placement in search/category results).
- Automatic vendor payouts: cash/COD orders paid out after 14 business days; other payment methods transferred automatically within 1 to 3 business days.
- An Account Health Score feature for each vendor, transparently showing their performance indicators.

3.4 Delivery

- Support for adding multiple delivery companies (such as FedEx and Aramex) within the system.
- If damage or an error occurs with a shipment, the responsible delivery company bears the cost — not a platform-funded protection fund.
- NEW: Real-time shipment tracking on an interactive map, with an accurate estimated time of arrival (ETA) that updates automatically based on the courier's location — not just a static "out for delivery" status.
- NEW: A delivery rating separate from the product rating (speed, package condition, courier conduct), to objectively monitor the performance of different delivery companies/couriers.

3.5 AI Customer Service

- A smart contact center/chatbot covering chat, WhatsApp, and voice calls together — one of Phase 1's top priorities.
- Automatic handling of return/exchange requests: understanding the issue, communicating with the vendor to resolve it, then replying to the customer — requiring only a final confirmation from the admin.
- AI trained on the entire system to automatically create short, quick tutorial videos (e.g. teaching vendors how to use the system).

3.6 Admin Panels

- Finance panel: accounting linked to sales, calculating each vendor's commission, calculating the platform's own cut, and clear, smart financial reports.
- Product approval panel: a simple review interface backed by AI alerts.

- An automatic background Audit Trail: logs all sensitive operations (commission edits, product approvals/rejections, permission changes, manual returns, admin logins) — who did what and when — to ensure transparency and accountability.
- No HR panel needed at this time.
- AI Fraud Detection — automatically monitors suspicious patterns (fake reviews, fake orders, attempts to abuse the points/referral system), alerting the admin when suspicious activity is detected.
- Compliant electronic tax invoicing (E-Invoice) — the system is ready to issue official invoices compliant with any future tax requirements (such as VAT) or B2B requirements.

3.7 Product Affiliate Marketing Program

- A dedicated section that lets vendors choose/opt in specific products of their own to participate in, as each vendor wishes.
- This section lets other people (affiliate marketers) select these available products and promote/sell them via their own link, in exchange for an additional commission when a sale is completed.

3.8 Vendor-Customer Chat

- Direct chat between the customer and vendor before completing a purchase (questions about size, availability, details, etc.) to reduce returns and increase trust, especially for products like clothing and jewelry.
- The chat is fully restricted and does not allow exchanging any personal data or contact information: sending phone numbers, emails, social media accounts, or any means of contact outside the platform is blocked — with automatic AI detection and filtering of any attempt to send this type of information, to prevent bypassing the platform for direct contact with the company.

3.9 Grocery / Supermarket Section

- The grocery page/section is built into the system now to be used later (not part of the actual first launch), because grocery products are displayed differently — some sold by count, others by kilo/weight/volume — similar to a supermarket like Whole Foods or local co-op societies.

4. Features Built Now, Activated Later

- A Kuwait-specific Loyalty Subscription — offering free fast delivery and exclusive deals, similar to "noon One" (not currently available in Kuwait).
- A Fully-Managed Fulfillment option: storage and shipping of products entirely handled by the platform on the vendor's behalf, especially in future integration with the Shiplee app.
- The Grocery/supermarket section — built into the architecture now, used later (see item 3.9).